

A Graduate-Level Reading Guide to *Optimal Counterfeiting Attacks and Generalizations for Wiesner's Quantum Money*

Molina, Vidick, and Watrous (2012)

Supplementary Notes for Quantum Information (M1-S2)

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1 Layer 0: The Big Picture — Why Quantum Money?

1.1 The classical counterfeiting problem

A banknote is, at bottom, a piece of information. If you can read that information perfectly, you can copy it. Classical information—strings of bits—can always be read and copied. Every anti-counterfeiting measure on a classical bill (watermarks, holograms, microprinting) is an attempt to make copying *expensive*, but never *impossible in principle*.

Quantum mechanics changes this. The **no-cloning theorem** (Wootters–Zurek, 1982) states:

Theorem 1.1 (No-Cloning). *There is no unitary operator U acting on two registers such that*

$$U |\psi\rangle |0\rangle = |\psi\rangle |\psi\rangle$$

for every quantum state $|\psi\rangle$.

Proof. Suppose such a U exists. Take two distinct nonorthogonal states $|\psi\rangle$ and $|\phi\rangle$ (so $0 < |\langle\psi|\phi\rangle| < 1$). By assumption:

$$U |\psi\rangle |0\rangle = |\psi\rangle |\psi\rangle, \quad U |\phi\rangle |0\rangle = |\phi\rangle |\phi\rangle.$$

Taking the inner product of both sides and using the fact that U is unitary ($U^\dagger U = \mathbb{1}$, so it preserves inner products):

$$\langle\psi|\phi\rangle \cdot \underbrace{\langle 0|0\rangle}_{=1} = (\langle\psi| \langle 0|) U^\dagger U (|\phi\rangle |0\rangle) = \langle\psi|\phi\rangle \cdot \langle\psi|\phi\rangle = \langle\psi|\phi\rangle^2.$$

Let $\alpha = \langle\psi|\phi\rangle$. Then $\alpha = \alpha^2$, so $\alpha(\alpha - 1) = 0$, giving $\alpha \in \{0, 1\}$. But we chose $0 < |\alpha| < 1$ —contradiction. $\square$

The physics behind no-cloning. The proof ultimately relies on the fact that quantum evolution is *linear*. A unitary U acts linearly on states: $U(\alpha|\psi\rangle + \beta|\phi\rangle) = \alpha U|\psi\rangle + \beta U|\phi\rangle$. If U clones $|0\rangle$ and $|1\rangle$ individually, then for $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$:

$$U|+\rangle|0\rangle = \frac{1}{\sqrt{2}}(U|0\rangle|0\rangle + U|1\rangle|0\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

But a proper clone of $|+\rangle$ should be $|+\rangle|+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$. Instead, U produces a *Bell state*—an entangled state, not two independent copies. Linearity forces this.

Key insight for the paper. No-cloning forbids *perfect* cloning of *all* states. But a counterfeiter does not need to clone all states—only the specific states used by the bank. And the counterfeiter does not need perfection—just a high enough success probability. The central question of this paper is: *how well can an optimal counterfeiter do?*

1.2 Wiesner’s original idea (1983)

Stephen Wiesner proposed “conjugate coding” around 1970 (published 1983). The scheme is:

1. **Preparation.** The bank picks, for each of n qubits in a banknote, a state uniformly at random from $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$. The choice is recorded secretly alongside a serial number.

2. **Verification.** To check a note, the bank measures each qubit in the basis that was used to prepare it (Z-basis for $|0\rangle, |1\rangle$; X-basis for $|+\rangle, |-\rangle$) and checks the outcomes match.

The four states come from two *conjugate bases*:

$$\text{Z-basis: } |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \text{X-basis: } |+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

These two bases are related by the Hadamard gate $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$: $H|0\rangle = |+\rangle$ and $H|1\rangle = |-\rangle$.

1.3 The information-disturbance tradeoff

This is the physical principle that makes quantum money work. Let us see it concretely.

Suppose a counterfeiter receives a qubit that is either $|0\rangle$ or $|+\rangle$, each with probability $1/2$. The counterfeiter wants to produce *two* copies of whatever state was sent. Consider the simplest strategy: measure, then prepare two copies of the outcome.

Strategy 1: Measure in Z-basis.

- If the state was $|0\rangle$: measurement gives 0 with certainty. Counterfeiter prepares $|0\rangle|0\rangle$. Both copies pass Z-basis verification. *Success*.
- If the state was $|+\rangle$: measurement gives 0 or 1 each with probability $1/2$. Say outcome is 0; counterfeiter prepares $|0\rangle|0\rangle$. But the bank verifies in the X-basis, measuring $|0\rangle$ and obtaining $+$ or $-$ each with probability $1/2$. Each copy passes with probability $1/2$, so both pass with probability $1/4$.

Overall success: $\frac{1}{2}(1) + \frac{1}{2}(\frac{1}{4}) = 5/8$.

Strategy 2: Measure in X-basis. By symmetry, this also gives success probability $5/8$.

The tension. Measuring in Z is perfect for Z-states but destroys X-states. Measuring in X is perfect for X-states but destroys Z-states. No single measurement can simultaneously preserve both. The optimal attack (as proven in this paper) achieves $3/4$ per qubit—better than either naive strategy, but still strictly less than 1.

This is a quantitative manifestation of the **Heisenberg uncertainty principle**: the Z- and X-bases are eigenbases of non-commuting observables (σ_z and σ_x), and precise knowledge of one precludes precise knowledge of the other.

1.4 What this paper proves

The main result (Theorem 1 of the paper) is:

Theorem 1.2 (Molina–Vidick–Watrous). *The optimal probability for a counterfeiter to produce two copies of an n -qubit Wiesner banknote, both passing verification, is exactly $(\frac{3}{4})^n$.*

For $n = 20$ qubits, this is roughly 3.2×10^{-3} . For $n = 100$, it is about 3.2×10^{-13} . Security is exponential in n .

The paper also proves results about better ensembles (the 6-state scheme achieves $(2/3)^n$), a classical-verification variant (success probability $(3/4 + \sqrt{2}/8)^n \approx (0.927)^n$), and higher-dimensional generalizations using qudits. All results are obtained through **semidefinite programming (SDP)**.

2 Layer 1: Quantum States, Measurements, and Channels

2.1 Pure states and the Bloch sphere

A qubit state $|\psi\rangle \in \mathbb{C}^2$ (up to global phase) can be written as

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle,$$

parametrized by $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$. This identifies the state with a point on the **Bloch sphere**, the unit sphere in $\mathbb{R}^3$, via the Bloch vector

$$\vec{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta).$$

The four Wiesner states live on this sphere:

$$\begin{array}{llll} |0\rangle: & (0, 0, 1) & |1\rangle: & (0, 0, -1) \\ |+\rangle: & (1, 0, 0) & |-\rangle: & (-1, 0, 0) \\ |i\rangle: & (0, 1, 0) & |-i\rangle: & (0, -1, 0) \end{array}$$

Inner product and geometry. For two pure states with Bloch vectors $\vec{r}_1$ and $\vec{r}_2$ separated by angle Θ on the sphere, the overlap is

$$|\langle \psi_1 | \psi_2 \rangle|^2 = \cos^2 \frac{\Theta}{2} = \frac{1 + \vec{r}_1 \cdot \vec{r}_2}{2} = \frac{1 + \cos \Theta}{2}.$$

This is a crucial formula. Two states are *orthogonal* ($\langle \psi_1 | \psi_2 \rangle = 0$) if and only if they are *antipodal* ($\Theta = \pi$) on the Bloch sphere. For $|0\rangle$ and $|+\rangle$, the angle is $\Theta = \pi/2$ (they are at 90°), giving $|\langle 0 | + \rangle|^2 = 1/2$.

Remark 2.1. The factor of $1/2$ in the parametrization $\theta/2$ means that orthogonal quantum states ($\langle \psi | \phi \rangle = 0$) are 180° apart on the Bloch sphere, not 90° . This doubling is related to the fact that $SU(2)$ is a double cover of $SO(3)$.

2.2 Density operators and the Bloch ball

When we have *classical* uncertainty about which pure state a system is in, we use a **density operator** ρ . If the system is in state $|\psi_k\rangle$ with probability p_k , then

$$\rho = \sum_k p_k |\psi_k\rangle \langle \psi_k|.$$

A density operator satisfies: (i) $\rho \geq 0$ (positive semidefinite), (ii) $\text{Tr}(\rho) = 1$. We write $D(\mathcal{X})$ for the set of all density operators on $\mathcal{X}$.

For a single qubit, every density operator can be written using the **Pauli matrices** $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ as

$$\rho = \frac{1}{2}(\mathbb{1} + \vec{r} \cdot \vec{\sigma}) = \frac{1}{2}(\mathbb{1} + r_x \sigma_x + r_y \sigma_y + r_z \sigma_z), \quad (1)$$

where $\vec{r} \in \mathbb{R}^3$ with $|\vec{r}| \leq 1$. Pure states satisfy $|\vec{r}| = 1$ (on the sphere), while mixed states have $|\vec{r}| < 1$ (inside the ball). The maximally mixed state $\rho = \mathbb{1}/2$ sits at the origin $\vec{r} = \vec{0}$.

Wiesner's ensemble average. For Wiesner's scheme, the bank's ensemble gives

$$\rho_{\text{avg}} = \frac{1}{4}(|0\rangle \langle 0| + |1\rangle \langle 1| + |+\rangle \langle +| + |-\rangle \langle -|) = \frac{1}{4}(\mathbb{1} + \mathbb{1}) = \frac{1}{2}\mathbb{1}.$$

The average state is maximally mixed: the Bloch vectors of the four states ($+\hat{z}$, $-\hat{z}$, $+\hat{x}$, $-\hat{x}$) cancel pairwise. This means that no measurement on a single qubit can reveal any information about the secret key.

2.3 Measurements (POVMs)

Definition 2.2. A *positive operator-valued measure (POVM)* on $\mathcal{X}$ is a set of positive semidefinite operators $\{P_a : a \in \Sigma\} \subset \text{Pos}(\mathcal{X})$, indexed by measurement outcomes Σ , satisfying $\sum_{a \in \Sigma} P_a = \mathbb{1}_{\mathcal{X}}$. When measuring state ρ , outcome a occurs with probability $\text{Tr}(P_a \rho)$.

Projective measurements are the special case where each P_a is a projector ($P_a^2 = P_a$) and the projectors are mutually orthogonal ($P_a P_b = \delta_{ab} P_a$). Measuring in the Z-basis means using $\{P_0 = |0\rangle\langle 0|, P_1 = |1\rangle\langle 1|\}$.

Example 2.3 (Why POVMs matter for counterfeiting). Suppose a counterfeiter receives a qubit uniformly from $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$ and must guess which state was sent. With a projective measurement (in any basis), the best strategy distinguishes between two of the four states perfectly but is random on the other two, achieving success probability $1/2$. But the optimal POVM has four elements (one “pointing toward” each state on the Bloch sphere) and achieves success probability $1/2 + 1/(2\sqrt{2}) \approx 0.854$. The extra freedom of having non-orthogonal measurement operators helps extract more information.

The counterfeiter in this paper similarly benefits from using general POVMs rather than projective measurements.

2.4 Quantum channels

Definition 2.4. A *quantum channel* is a linear map $\Phi : L(\mathcal{X}) \rightarrow L(\mathcal{Y})$ that is:

1. **Completely positive (CP):** For every ancilla space $\mathcal{A}$ of any dimension, the extended map $\Phi \otimes \text{id}_{\mathcal{A}}$ sends positive semidefinite operators to positive semidefinite operators.
2. **Trace-preserving (TP):** $\text{Tr}(\Phi(\rho)) = \text{Tr}(\rho)$ for all ρ .

Why “completely” positive, not just positive? A map Φ is *positive* if $\rho \geq 0 \Rightarrow \Phi(\rho) \geq 0$. But in quantum mechanics, the system may be entangled with an ancilla. We need $(\Phi \otimes \text{id})(\rho_{XA}) \geq 0$ for every joint state ρ_{XA} , even entangled ones. Positive but not completely positive maps exist (the transpose map $\rho \mapsto \rho^T$ is the canonical example), and they are not physical. The Choi–Jamiołkowski isomorphism (Section 3) gives a clean test: Φ is CP if and only if $J(\Phi) \geq 0$.

Kraus decomposition. Every quantum channel has a **Kraus decomposition**:

$$\Phi(\rho) = \sum_{i=1}^r A_i \rho A_i^\dagger, \quad \text{with} \quad \sum_{i=1}^r A_i^\dagger A_i = \mathbb{1}_{\mathcal{X}}.$$

The operators $A_i : \mathcal{X} \rightarrow \mathcal{Y}$ are called **Kraus operators**. The condition $\sum_i A_i^\dagger A_i = \mathbb{1}$ enforces trace preservation. The number r of Kraus operators is the **Kraus rank**; $r = 1$ corresponds to a unitary channel ($\Phi(\rho) = U\rho U^\dagger$).

Physical picture. A Kraus decomposition can be interpreted as follows: the system interacts with an environment, and the environment is then discarded (traced out). Each Kraus operator A_i corresponds to one possible “branch” of the interaction. This is equivalent to the **Stinespring dilation**: there exists an isometry $V : \mathcal{X} \rightarrow \mathcal{Y} \otimes \mathcal{E}$ such that $\Phi(\rho) = \text{Tr}_{\mathcal{E}}(V\rho V^\dagger)$, with V related to the A_i by choosing an orthonormal basis $\{|i\rangle\}$ of $\mathcal{E}$ and setting $V = \sum_i A_i \otimes |i\rangle$.

Counterfeiting as a channel. A counterfeiting attack is a channel $\Phi : L(\mathcal{X}) \rightarrow L(\mathcal{Y} \otimes \mathcal{Z})$ taking one register (the banknote, dimension d) to two registers (the two “copies”, each of dimension d). The Kraus operators are $A_i : \mathbb{C}^d \rightarrow \mathbb{C}^d \otimes \mathbb{C}^d$, i.e., $d^2 \times d$ matrices.

3 Layer 2: The Choi–Jamiołkowski Isomorphism

This is the central tool that converts optimization over *channels* into optimization over *matrices*.

3.1 The maximally entangled state

Fix an orthonormal basis $\{|1\rangle, \dots, |d\rangle\}$ of $\mathcal{X}$ with $d = \dim(\mathcal{X})$. The (unnormalized) **maximally entangled state** is

$$|\Omega\rangle = \sum_{i=1}^d |i\rangle |i\rangle \in \mathcal{X} \otimes \mathcal{X}.$$

(The normalized version is $|\Omega\rangle / \sqrt{d}$.) This state has a remarkable property: for any operator $A \in L(\mathcal{X})$,

$$(A \otimes \mathbb{1}) |\Omega\rangle = (\mathbb{1} \otimes A^T) |\Omega\rangle, \quad (2)$$

where A^T is the transpose of A in the chosen basis. This identity is sometimes called the “transpose trick” or “ricochet property.”

3.2 Definition of the CJ operator

Definition 3.1. *The Choi–Jamiołkowski (CJ) operator of a linear map $\Phi : L(\mathcal{X}) \rightarrow L(\mathcal{Y})$ is*

$$J(\Phi) = \sum_{i,j=1}^d \Phi(|i\rangle \langle j|) \otimes |i\rangle \langle j| = (\Phi \otimes \text{id})(|\Omega\rangle \langle \Omega|) \in L(\mathcal{Y} \otimes \mathcal{X}).$$

The second equality shows that $J(\Phi)$ is obtained by applying Φ to one half of the maximally entangled state $|\Omega\rangle$. This is why the CJ operator is sometimes called the “Choi state” (up to normalization) of the channel.

Theorem 3.2 (Choi 1975, Jamiołkowski 1972). *The map $\Phi \mapsto J(\Phi)$ is a linear bijection. Moreover:*

1. Φ is completely positive $\iff J(\Phi) \geq 0$.
2. Φ is trace-preserving $\iff \text{Tr}_{\mathcal{Y}}(J(\Phi)) = \mathbb{1}_{\mathcal{X}}$.

Therefore, Φ is a quantum channel $\iff J(\Phi) \geq 0$ and $\text{Tr}_{\mathcal{Y}}(J(\Phi)) = \mathbb{1}_{\mathcal{X}}$.

3.3 The key identity and the role of complex conjugation

For any $|\psi\rangle \in \mathcal{X}$ and $|\phi\rangle \in \mathcal{Y}$:

$$\langle \phi | \Phi(|\psi\rangle \langle \psi|) | \phi \rangle = \langle \phi \otimes \bar{\psi} | J(\Phi) | \phi \otimes \bar{\psi} \rangle, \quad (3)$$

where $\bar{\psi}$ denotes the **entry-wise complex conjugate** of $|\psi\rangle$ in the chosen basis: if $|\psi\rangle = \sum_i \alpha_i |i\rangle$, then $|\bar{\psi}\rangle = \sum_i \alpha_i^* |i\rangle$.

Where does $\bar{\psi}$ come from? The proof uses the transpose trick (2). Expanding:

$$\begin{aligned}
\langle \phi \otimes \bar{\psi} | J(\Phi) | \phi \otimes \bar{\psi} \rangle &= \langle \phi \otimes \bar{\psi} | (\Phi \otimes \text{id})(|\Omega\rangle \langle \Omega|) | \phi \otimes \bar{\psi} \rangle \\
&= \sum_{i,j} \langle \phi | \Phi(|i\rangle \langle j|) | \phi \rangle \cdot \langle \bar{\psi} | i \rangle \langle j | \bar{\psi} \rangle \\
&= \sum_{i,j} \langle \phi | \Phi(|i\rangle \langle j|) | \phi \rangle \cdot \psi_i \cdot \psi_j^* \\
&= \langle \phi | \Phi \left(\sum_{i,j} \psi_i \psi_j^* |i\rangle \langle j| \right) | \phi \rangle \\
&= \langle \phi | \Phi(|\psi\rangle \langle \psi|) | \phi \rangle.
\end{aligned}$$

The key step: $\langle \bar{\psi} | i \rangle = \psi_i$ (not ψ_i^*), because the complex conjugate swaps the roles of ψ_i and ψ_i^* relative to the bra-ket convention.

Remark 3.3. For states with real amplitudes in the standard basis (like $|0\rangle, |1\rangle, |+\rangle, |-\rangle$), we have $|\bar{\psi}\rangle = |\psi\rangle$, so the conjugate makes no difference. It only matters for states with complex phases, like $|+i\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$, for which $|\bar{+i}\rangle = (|0\rangle - i|1\rangle)/\sqrt{2} = |-i\rangle$.

3.4 A worked example

Consider the depolarizing channel on a single qubit:

$$\Phi(\rho) = (1-p)\rho + p \frac{\mathbb{1}}{2}, \quad 0 \leq p \leq 1.$$

To compute $J(\Phi)$, apply Φ to one half of $|\Omega\rangle \langle \Omega|$ where $|\Omega\rangle = |00\rangle + |11\rangle$:

$$|\Omega\rangle \langle \Omega| = |00\rangle \langle 00| + |00\rangle \langle 11| + |11\rangle \langle 00| + |11\rangle \langle 11|.$$

Applying Φ to the first register:

$$\begin{aligned}
J(\Phi) &= (1-p) |\Omega\rangle \langle \Omega| + p \frac{\mathbb{1}}{2} \otimes (|0\rangle \langle 0| + |1\rangle \langle 1|) \\
&= (1-p) |\Omega\rangle \langle \Omega| + \frac{p}{2} \mathbb{1}_4.
\end{aligned}$$

You can verify: $J(\Phi) \geq 0$ (eigenvalues are $1 - 3p/4$ and $p/4$ each with multiplicity 3), and $\text{Tr}_Y(J(\Phi)) = \mathbb{1}_2$ (trace-preserving). At $p = 0$ (no noise), $J(\Phi) = |\Omega\rangle \langle \Omega|$, a rank-1 projector. At $p = 1$ (total depolarization), $J(\Phi) = \mathbb{1}_4/2$, proportional to the identity.

3.5 Three equivalent representations

Representation	Form	Advantage
Kraus	$\Phi(\rho) = \sum_i A_i \rho A_i^\dagger$	Physical interpretation
Stinespring	$\Phi(\rho) = \text{Tr}_E(V \rho V^\dagger)$	Unitary dilation
Choi-Jamiołkowski	$J(\Phi) \geq 0, \text{Tr}_Y J(\Phi) = \mathbb{1}$	SDP-compatible

The CJ form is preferred for optimization because the set of all channels becomes a convex set defined by semidefinite constraints.

4 Layer 3: Semidefinite Programming (SDP)

4.1 From linear programming to SDPs

If you know linear programming (LP) from optimization or CS courses:

	LP	SDP
Variable	$x \in \mathbb{R}^n$	$X \in \text{Herm}(\mathcal{X})$
Objective	$c^T x$	$\langle A, X \rangle = \text{Tr}(A^\dagger X)$
Constraint	$Ax = b, x \geq 0$	$\Phi(X) = B, X \geq 0$ (PSD)

SDPs generalize LPs by replacing the vector variable $x \geq 0$ (componentwise) with a matrix variable $X \geq 0$ (positive semidefinite). The condition $X \geq 0$ defines a cone—the positive semidefinite cone—which is the matrix analogue of the nonnegative orthant.

4.2 Formal definition

A semidefinite program is a triple (Φ, A, B) where $\Phi : L(\mathcal{X}) \rightarrow L(\mathcal{Y})$ is a Hermiticity-preserving linear map, $A \in \text{Herm}(\mathcal{X})$, $B \in \text{Herm}(\mathcal{Y})$. The associated optimization problems are:

<u>Primal problem</u>	<u>Dual problem</u>
maximize: $\langle A, X \rangle$	minimize: $\langle B, Y \rangle$
subject to: $\Phi(X) = B$	subject to: $\Phi^*(Y) \geq A$
$X \geq 0$	$Y \in \text{Herm}(\mathcal{Y})$

The **adjoint** Φ^* is the unique map satisfying $\langle Y, \Phi(X) \rangle = \langle \Phi^*(Y), X \rangle$ for all X, Y . In terms of the Hilbert–Schmidt inner product $\langle A, B \rangle = \text{Tr}(A^\dagger B)$, this means $\text{Tr}(Y^\dagger \Phi(X)) = \text{Tr}(\Phi^*(Y)^\dagger X)$ for all $X \in L(\mathcal{X})$, $Y \in L(\mathcal{Y})$.

4.3 Weak duality

Theorem 4.1 (Weak duality). *If X is primal feasible and Y is dual feasible, then $\langle A, X \rangle \leq \langle B, Y \rangle$.*

Proof. Since $\Phi^*(Y) \geq A$ and $X \geq 0$:

$$\langle A, X \rangle \leq \langle \Phi^*(Y), X \rangle = \langle Y, \Phi(X) \rangle = \langle Y, B \rangle = \langle B, Y \rangle.$$

The first inequality uses: if $M \geq N$ and $X \geq 0$, then $\text{Tr}(MX) \geq \text{Tr}(NX)$ (take the spectral decomposition of X). The second step uses the definition of the adjoint. The third uses $\Phi(X) = B$. $\square$

Consequence: Any feasible dual solution gives an *upper bound* on the primal optimum. This is the key tool for proving security: if we find a dual feasible Y with $\langle B, Y \rangle = 3/4$, then no counterfeiting strategy can exceed probability $3/4$.

4.4 Strong duality (Slater’s theorem)

Theorem 4.2 (Slater’s theorem). *If the dual optimum β is finite and there exists a strictly feasible primal point ($X > 0$ with $\Phi(X) = B$), then $\alpha = \beta$ and the dual optimum is attained (and vice versa with roles swapped).*

Strong duality lets us prove *exact* optimality: if we exhibit primal and dual solutions with the same objective value, both must be optimal.

5 Layer 4: The Counterfeiting SDP

5.1 Setting up the optimization

We have an ensemble $\mathcal{E} = \{(p_k, |\psi_k\rangle) : k = 1, \dots, N\}$ and a counterfeiting channel $\Phi : L(\mathcal{X}) \rightarrow L(\mathcal{Y} \otimes \mathcal{Z})$. The success probability is

$$p_{\text{success}} = \sum_{k=1}^N p_k \langle \psi_k \otimes \psi_k | \Phi(|\psi_k\rangle \langle \psi_k|) | \psi_k \otimes \psi_k \rangle.$$

Using the CJ identity (3) with $|\phi\rangle = |\psi_k \otimes \psi_k\rangle \in \mathcal{Y} \otimes \mathcal{Z}$:

$$\begin{aligned} p_{\text{success}} &= \sum_{k=1}^N p_k \langle \psi_k \otimes \psi_k \otimes \bar{\psi}_k | J(\Phi) | \psi_k \otimes \psi_k \otimes \bar{\psi}_k \rangle \\ &= \text{Tr} \left(\underbrace{\left[\sum_{k=1}^N p_k | \psi_k \otimes \psi_k \otimes \bar{\psi}_k \rangle \langle \psi_k \otimes \psi_k \otimes \bar{\psi}_k | \right]}_{=Q} \cdot J(\Phi) \right) \\ &= \langle Q, J(\Phi) \rangle. \end{aligned}$$

So the success probability is a *linear function* of $X = J(\Phi)$.

5.2 Identifying the SDP structure

Setting $X = J(\Phi) \in L(\mathcal{Y} \otimes \mathcal{Z} \otimes \mathcal{X})$:

1. **Objective:** $\langle Q, X \rangle$ (linear in X , with Q defined above).
2. **CP constraint:** $X \geq 0$ (positive semidefinite).
3. **TP constraint:** $\text{Tr}_{\mathcal{Y} \otimes \mathcal{Z}}(X) = \mathbb{1}_{\mathcal{X}}$.

This is exactly the SDP:

Primal (counterfeiter's optimization):

$$\alpha = \max \{ \langle Q, X \rangle : \text{Tr}_{\mathcal{Y} \otimes \mathcal{Z}}(X) = \mathbb{1}_{\mathcal{X}}, X \geq 0 \}.$$

To derive the dual, we identify: $A = Q$, $B = \mathbb{1}_{\mathcal{X}}$, and $\Phi(X) = \text{Tr}_{\mathcal{Y} \otimes \mathcal{Z}}(X)$. The adjoint of the partial trace is the *tensoring with identity*: for any $Y \in \text{Herm}(\mathcal{X})$,

$$\Phi^*(Y) = \mathbb{1}_{\mathcal{Y} \otimes \mathcal{Z}} \otimes Y.$$

(To verify: $\langle Y, \Phi(X) \rangle = \text{Tr}(Y \cdot \text{Tr}_{\mathcal{Y} \otimes \mathcal{Z}}(X)) = \text{Tr}((\mathbb{1}_{\mathcal{Y} \otimes \mathcal{Z}} \otimes Y) \cdot X) = \langle \Phi^*(Y), X \rangle$.)

The dual constraint $\Phi^*(Y) \geq A$ becomes $\mathbb{1}_{\mathcal{Y} \otimes \mathcal{Z}} \otimes Y \geq Q$, and $\langle B, Y \rangle = \text{Tr}(\mathbb{1}_{\mathcal{X}} \cdot Y) = \text{Tr}(Y)$.

Dual (security certificate):

$$\beta = \min \{ \text{Tr}(Y) : \mathbb{1}_{\mathcal{Y} \otimes \mathcal{Z}} \otimes Y \geq Q, Y \in \text{Herm}(\mathcal{X}) \}.$$

Remark 5.1 (Strict feasibility). *The primal is strictly feasible: $X = \frac{1}{d^2} \mathbb{1}_{\mathcal{Y} \otimes \mathcal{Z} \otimes \mathcal{X}}$ satisfies $X > 0$ and $\text{Tr}_{\mathcal{Y} \otimes \mathcal{Z}}(X) = \mathbb{1}_{\mathcal{X}}$. Similarly, $Y = \lambda \mathbb{1}$ for large λ satisfies the dual constraint strictly. By Slater's theorem, strong duality holds.*

5.3 The dual shortcut via the operator norm

If we guess the dual solution $Y = \mu \mathbb{1}_{\mathcal{X}}$ for some scalar μ , the dual constraint becomes $\mu \mathbb{1}_{\mathcal{Y} \otimes \mathcal{Z} \otimes \mathcal{X}} \geq Q$, which holds if and only if $\mu \geq \|Q\|$ (the operator norm, i.e., the largest eigenvalue of Q). The dual objective is then $\text{Tr}(Y) = \mu d$ where $d = \dim(\mathcal{X})$. The best such choice is $\mu = \|Q\|$, giving:

$$\alpha \leq \beta \leq d \cdot \|Q\|. \quad (4)$$

If we can also exhibit a primal feasible X achieving this same value, we have proven $\alpha = d \cdot \|Q\|$.

6 Layer 5: Solving the Wiesner SDP

6.1 Computing Q for Wiesner's 4-state ensemble

The ensemble is $\mathcal{E} = \{(\frac{1}{4}, |0\rangle), (\frac{1}{4}, |1\rangle), (\frac{1}{4}, |+\rangle), (\frac{1}{4}, |-\rangle)\}$. Since all four states have real amplitudes, $|\overline{\psi_k}\rangle = |\psi_k\rangle$, so:

$$Q = \frac{1}{4}(|000\rangle\langle 000| + |111\rangle\langle 111| + |+++ \rangle\langle +++| + |-- \rangle\langle --|).$$

Here the three tensor factors correspond to registers $\mathcal{Y}, \mathcal{Z}, \mathcal{X}$.

6.2 Computing $\|Q\| = 3/8$: the eigenvalue analysis

The operator Q acts on $(\mathbb{C}^2)^{\otimes 3} \cong \mathbb{C}^8$. To find its largest eigenvalue, we exploit the symmetry of the ensemble.

Step 1: Expand the $+/-$ states.

$$\begin{aligned} |+++ \rangle &= \frac{1}{2\sqrt{2}}(|0\rangle + |1\rangle)(|0\rangle + |1\rangle)(|0\rangle + |1\rangle), \\ |-- \rangle &= \frac{1}{2\sqrt{2}}(|0\rangle - |1\rangle)(|0\rangle - |1\rangle)(|0\rangle - |1\rangle). \end{aligned}$$

Step 2: Combine contributions. Let $P_{000} = |000\rangle\langle 000|$ and $P_{111} = |111\rangle\langle 111|$. The “computational basis” part of Q is $\frac{1}{4}(P_{000} + P_{111})$, contributing rank-1 projectors along $|000\rangle$ and $|111\rangle$.

The “Hadamard basis” part $\frac{1}{4}(|+++ \rangle\langle +++| + |-- \rangle\langle --|)$ spreads weight across all 8 basis states, but the cross-terms between $|0\rangle^{\otimes 3}$ and $|1\rangle^{\otimes 3}$ cancel (because $|+\rangle$ and $|-\rangle$ contribute opposite signs for terms with odd parity).

Step 3: Identify the top eigenvalue. By the symmetry of Q under the permutation of the $\mathcal{Y}$ and $\mathcal{Z}$ registers, the eigenvectors of Q can be found within the symmetric and antisymmetric subspaces. A direct computation (or numerical diagonalization of the 8×8 matrix) yields the spectrum:

$$\text{eigenvalues of } Q: \quad \frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, 0, 0, 0, 0.$$

The top eigenvector is $(|000\rangle + |+++ \rangle)/\sqrt{2}$ (normalization), and the largest eigenvalue is $\|Q\| = 3/8$.

Alternative: direct verification. Even without computing the full spectrum, one can verify that $\|Q\| \leq 3/8$ by checking that $\frac{3}{8}\mathbb{1} - Q \geq 0$ (all eigenvalues of $\frac{3}{8}\mathbb{1} - Q$ are nonnegative). Then the dual solution $Y = \frac{3}{8}\mathbb{1}_{\mathcal{X}}$ gives objective value $\text{Tr}(Y) = 2 \cdot \frac{3}{8} = 3/4$.

6.3 The optimal attack (primal solution)

To show $3/4$ is achievable, we exhibit a channel $\Phi(\rho) = A_0\rho A_0^\dagger + A_1\rho A_1^\dagger$ with:

$$A_0 = \frac{1}{\sqrt{12}} \begin{pmatrix} 3 & 0 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad A_1 = \frac{1}{\sqrt{12}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ 0 & 3 \end{pmatrix}.$$

These are 4×2 matrices mapping $\mathbb{C}^2 \rightarrow \mathbb{C}^4 \cong \mathbb{C}^2 \otimes \mathbb{C}^2$.

Verification 1: Trace preservation.

$$A_0^\dagger A_0 = \frac{1}{12} \begin{pmatrix} 3 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{1}{12} \begin{pmatrix} 10 & 0 \\ 0 & 2 \end{pmatrix},$$

$$A_1^\dagger A_1 = \frac{1}{12} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ 0 & 3 \end{pmatrix} = \frac{1}{12} \begin{pmatrix} 2 & 0 \\ 0 & 10 \end{pmatrix}.$$

Sum: $A_0^\dagger A_0 + A_1^\dagger A_1 = \frac{1}{12} \begin{pmatrix} 12 & 0 \\ 0 & 12 \end{pmatrix} = \mathbb{1}_2$. ✓

Verification 2: Success probability for $|0\rangle$.

$$A_0 |0\rangle = \frac{1}{\sqrt{12}} (3, 0, 0, 1)^T, \quad A_1 |0\rangle = \frac{1}{\sqrt{12}} (0, 1, 1, 0)^T.$$

$$\Phi(|0\rangle\langle 0|) = \frac{1}{12} \begin{pmatrix} 9 & 0 & 0 & 3 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 3 & 0 & 0 & 1 \end{pmatrix} + \frac{1}{12} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \frac{1}{12} \begin{pmatrix} 9 & 0 & 0 & 3 \\ 0 & 2 & 2 & 0 \\ 0 & 2 & 2 & 0 \\ 3 & 0 & 0 & 1 \end{pmatrix}.$$

The success probability is the $(00,00)$ -entry of $\Phi(|0\rangle\langle 0|)$, which represents $\langle 00 | \Phi(|0\rangle\langle 0|) | 00 \rangle$:

$$\langle 00 | \Phi(|0\rangle\langle 0|) | 00 \rangle = \frac{9}{12} = \frac{3}{4}. \quad \checkmark$$

By the symmetry of A_0 and A_1 , the same holds for the other three states.

6.4 Intuition: what does the optimal attack do?

Consider input $|0\rangle$. The ideal clone would be $|00\rangle$. The channel produces a state with amplitude $9/12 = 3/4$ on $|00\rangle$, but also nonzero amplitudes on $|01\rangle + |10\rangle$ (the antisymmetric component is absent by the channel's symmetry). The attack *biases toward getting both copies right*, at the cost of sometimes getting *both wrong together*.

The $1/4$ failure probability comes from the information-disturbance tradeoff of Section 1.3. The channel must simultaneously work well for both Z-basis and X-basis inputs. The Kraus operators encode a compromise: A_0 “favors” $|0\rangle$ (large top-left entry 3), while A_1 favors $|1\rangle$ (large bottom-right entry 3). The off-diagonal entries (1's) allow partial success on the cross-basis states.

Connection to optimal cloning. This channel is closely related to the phase-covariant cloner studied by Bruß et al. (2000). It is the optimal cloner for the specific set $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$, but *not* a universal cloner (its performance varies by input state on the Bloch sphere).

7 Layer 6: Parallel Repetition and n -Qubit Security

7.1 From one qubit to n qubits

The n -qubit scheme independently prepares each qubit. The bank note state is $|\psi_{k_1}\rangle \otimes \cdots \otimes |\psi_{k_n}\rangle$ for independently chosen keys.

The natural mathematical structure is: the n -qubit problem has

$$Q^{(n)} = W_\pi (Q^{\otimes n}) W_\pi^*,$$

where Q is the single-qubit operator and W_π is a permutation unitary that rearranges tensor factors.

7.2 The permutation W_π explained

Each qubit repetition involves three registers $(\mathcal{Y}_j, \mathcal{Z}_j, \mathcal{X}_j)$. The natural tensor structure is $(\mathcal{Y}_1 \mathcal{Z}_1 \mathcal{X}_1) \otimes \cdots \otimes (\mathcal{Y}_n \mathcal{Z}_n \mathcal{X}_n)$, but the SDP variable lives in $(\mathcal{Y}_1 \cdots \mathcal{Y}_n) \otimes (\mathcal{Z}_1 \cdots \mathcal{Z}_n) \otimes (\mathcal{X}_1 \cdots \mathcal{X}_n)$. The unitary W_π simply reorders the factors:

$$W_\pi : |y_1 z_1 x_1\rangle \otimes \cdots \otimes |y_n z_n x_n\rangle \mapsto |y_1 \cdots y_n\rangle \otimes |z_1 \cdots z_n\rangle \otimes |x_1 \cdots x_n\rangle.$$

This is a permutation of subsystems, not a change of basis within any subsystem.

7.3 The product property of SDPs

The key result (Mittal–Szegedy 2007): for SDPs of the form in Section 5, when $Q \geq 0$ (which is automatic since Q is a sum of rank-1 PSD operators), the n -fold repetition has optimal value α^n .

Theorem 7.1 (Product property). *If the single-qubit SDP has optimal value α , then the n -qubit SDP has optimal value α^n .*

Proof sketch. Primal: α^n is achievable. Let X_1^* be the optimal single-qubit primal solution. Then $X = W_\pi (X_1^* \otimes \cdots \otimes X_1^*) W_\pi^*$ is feasible for the n -qubit SDP (tensor products of channels are channels), and achieves value α^n .

Dual: α^n cannot be exceeded. Let Y_1^* be the optimal single-qubit dual solution. Then $Y = Y_1^* \otimes \cdots \otimes Y_1^*$ is feasible for the n -qubit dual, achieving value $(\text{Tr}(Y_1^*))^n = \alpha^n$. Dual feasibility requires:

$$\mathbb{1} \otimes (Y_1^* \otimes \cdots \otimes Y_1^*) \geq W_\pi (Q^{\otimes n}) W_\pi^*.$$

This follows from the general fact below applied to $A = \mathbb{1} \otimes Y_1^*$ and $B = Q$. □

Lemma 7.2 (Tensor monotonicity of PSD ordering). *If $A \geq B \geq 0$ (both positive semidefinite, $A - B$ positive semidefinite), then $A^{\otimes n} \geq B^{\otimes n}$.*

Proof. For $n = 2$: write $A = B + C$ where $C = A - B \geq 0$. Then

$$A \otimes A = B \otimes B + B \otimes C + C \otimes B + C \otimes C.$$

Each of $B \otimes C, C \otimes B, C \otimes C$ is PSD (tensor product of PSD operators is PSD), so $A \otimes A \geq B \otimes B$. Induction gives the general case. □

Key consequence: A counterfeiter gains *nothing* by correlating qubits during an attack. Attacking each qubit independently and optimally is globally optimal.

7.4 Noise-tolerant (threshold) schemes

In practice, the bank might accept a note as valid if at least t out of n qubits pass, rather than requiring all n . If the single-qubit scheme satisfies:

1. $\sum_k p_k |\psi_k\rangle \langle \psi_k| = \frac{1}{d} \mathbb{1}$ (maximally mixed ensemble average), and
2. $Y = \frac{\alpha}{d} \mathbb{1}$ is an optimal dual solution,

then the optimal counterfeiting probability for the threshold- t scheme is:

$$p_{\text{success}}^{(t,n)} = \sum_{j=t}^n \binom{n}{j} \alpha^j (1-\alpha)^{n-j}.$$

This is exactly the probability of $\geq t$ successes in n independent Bernoulli trials with success probability α . Both Wiesner's scheme and the schemes in the paper satisfy these conditions.

8 Layer 7: Better Schemes and the Role of Symmetry

8.1 The symmetric subspace

Before discussing better schemes, we need a key concept.

Definition 8.1. The *symmetric subspace* $\text{Sym}^k(\mathbb{C}^d)$ of $(\mathbb{C}^d)^{\otimes k}$ is the subspace of vectors invariant under all permutations of the k tensor factors:

$$\text{Sym}^k(\mathbb{C}^d) = \{ |v\rangle \in (\mathbb{C}^d)^{\otimes k} : P_\sigma |v\rangle = |v\rangle \text{ for all } \sigma \in S_k \},$$

where P_σ permutes the tensor factors according to σ .

Its dimension is $\binom{d+k-1}{k}$. For $k = 3$ qubits ($d = 2$): $\dim(\text{Sym}^3(\mathbb{C}^2)) = \binom{3}{3} = 4$. A basis for this 4-dimensional subspace is:

$$|000\rangle, \quad \frac{|001\rangle + |010\rangle + |100\rangle}{\sqrt{3}}, \quad \frac{|011\rangle + |101\rangle + |110\rangle}{\sqrt{3}}, \quad |111\rangle.$$

The antisymmetric subspace is 0-dimensional for $k > d$, and for $k = 3, d = 2$, the remaining 4 dimensions form the complement.

The **projector onto the symmetric subspace** is $\Pi_{\text{sym}} = \frac{1}{k!} \sum_{\sigma \in S_k} P_\sigma$.

8.2 The 6-state scheme: $(2/3)^n$ security

Pastawski et al. proposed using all three Pauli eigenbases:

$$\mathcal{E}_6 = \{ \frac{1}{6} |0\rangle, \frac{1}{6} |1\rangle, \frac{1}{6} |+\rangle, \frac{1}{6} |-\rangle, \frac{1}{6} |+i\rangle, \frac{1}{6} |-i\rangle \},$$

where $|\pm i\rangle = (|0\rangle \pm i|1\rangle)/\sqrt{2}$ (eigenstates of σ_y , lying along the $\pm y$ axis of the Bloch sphere).

A remarkable fact is that the resulting Q operator takes the form:

$$Q = \frac{1}{\text{rank}(\Pi_{\text{sym}})} (\text{id} \otimes \text{id} \otimes T) (\Pi_{\text{sym}}), \quad (5)$$

where T is the transposition map on the third register and Π_{sym} is the projector onto $\text{Sym}^3(\mathbb{C}^2)$.

Why does the symmetric subspace appear? The 6-state ensemble has the property that $\sum_k p_k |\psi_k\rangle \langle \psi_k|^{\otimes 3}$ reproduces the third moment of the uniform (Haar) distribution on the Bloch sphere. By Schur–Weyl duality, such averages project onto the symmetric subspace. (This is the definition of a quantum 3-design.)

The operator norm is $\|Q\| = 1/3$, giving dual bound $\text{Tr}(Y) = 2 \cdot 1/3 = 2/3$. The optimal attack is the **Bužek–Hillery universal cloner** (1996), with Kraus operators:

$$A_0 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad A_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

This cloner achieves $\langle \psi \otimes \psi | \Phi(|\psi\rangle \langle \psi|) | \psi \otimes \psi \rangle = 2/3$ for *every* state $|\psi\rangle$ on the Bloch sphere—it is universal.

8.3 SIC-POVM schemes

Definition 8.2. A **SIC-POVM** (symmetric informationally complete POVM) in dimension d is a set of d^2 rank-1 projectors $\{|\tau_j\rangle \langle \tau_j|\}$ satisfying $|\langle \tau_j | \tau_k \rangle|^2 = \frac{1}{d+1}$ for $j \neq k$.

For $d = 2$ (qubits), a SIC-POVM consists of 4 states forming a **regular tetrahedron** inscribed in the Bloch sphere. One explicit choice:

$$\begin{aligned} |\tau_1\rangle &= |0\rangle, \\ |\tau_2\rangle &= \frac{1}{\sqrt{3}} |0\rangle + \sqrt{\frac{2}{3}} |1\rangle, \\ |\tau_3\rangle &= \frac{1}{\sqrt{3}} |0\rangle + \sqrt{\frac{2}{3}} e^{2\pi i/3} |1\rangle, \\ |\tau_4\rangle &= \frac{1}{\sqrt{3}} |0\rangle + \sqrt{\frac{2}{3}} e^{4\pi i/3} |1\rangle. \end{aligned}$$

These satisfy $|\langle \tau_j | \tau_k \rangle|^2 = 1/3$ for $j \neq k$.

The ensemble $\mathcal{E} = \{(1/4, |\tau_j\rangle)\} : j = 1, \dots, 4\}$ yields the *same* Q operator as the 6-state scheme (5) (because the SIC-POVM also forms a quantum 3-design in dimension 2), and therefore also achieves optimal counterfeiting probability $2/3$.

Geometric intuition. The tetrahedron is the most “spread out” configuration of 4 points on the sphere. This gives the counterfeiter the least possible information about which state was chosen, making cloning as hard as possible.

8.4 Fundamental limits: Werner’s cloner

For any d -dimensional scheme, Werner’s universal cloner (1998) achieves counterfeiting probability $2/(d+1)$:

$$\langle \psi \otimes \psi | \Phi_{\text{Werner}}(|\psi\rangle \langle \psi|) | \psi \otimes \psi \rangle = \frac{2}{d+1} \quad \text{for all } |\psi\rangle \in \mathbb{C}^d.$$

This is a universal lower bound on counterfeiting probability: no scheme can do better than $(2/(d+1))^n$.

The paper shows this bound is *achievable* using ensembles from quantum 3-designs. The proof uses the same dual technique: compute $\|Q\| = 2/(d(d+1))$, so $\text{Tr}(Y) = d \cdot 2/(d(d+1)) = 2/(d+1)$.

Definition 8.3. A *quantum t -design* is a set of states $\{(p_k, |\psi_k\rangle)\}$ such that $\sum_k p_k (|\psi_k\rangle \langle \psi_k|)^{\otimes t}$ equals the t -th moment of the Haar measure on the unit sphere in $\mathbb{C}^d$. For $t = 3$, this means the ensemble “looks like” the uniform distribution up to third-order statistics.

9 Layer 8: Classical Verification — Quantum Tickets

9.1 Motivation: avoiding quantum communication

Wiesner’s original scheme requires the bank to physically receive and measure the qubits — *quantum communication*. This is technologically expensive. The classical-verification variant asks: can we verify banknotes using only *classical messages* back and forth, while the quantum state stays with the user?

The trade-off is clear: the bank learns less about the note (it only sees classical measurement outcomes, not the full quantum state), so verification is weaker, and counterfeiting is easier. But exponential security can still be achieved.

9.2 The ticket protocol

1. The bank prepares an n -qubit state as in Wiesner’s scheme.
2. To verify, the bank sends a random challenge $c \in \{0,1\}^n$: for each qubit i , $c_i = 0$ means “measure in Z -basis” and $c_i = 1$ means “measure in X -basis.”
3. The user measures each qubit in the requested basis and sends outcomes $b \in \{0,1\}^n$ to the bank.
4. The bank accepts if, whenever c_i matches the actual encoding basis of qubit i , the outcome b_i is correct. When c_i does *not* match, any answer is accepted.

Why any answer is accepted on mismatched bases. If qubit i was prepared in the Z -basis (say $|0\rangle$) but the challenge asks for an X -basis measurement, even an honest user gets a random outcome (+ or – with equal probability). The bank cannot tell whether the user answered correctly, so it accepts any answer.

9.3 The counterfeiting problem for tickets

A counterfeiter receives one ticket and must answer *two* independent challenges $c_1, c_2 \in \{0,1\}$ from the bank. (We analyze the single-qubit case; n qubits follow by parallel repetition.)

Case 1: $c_1 = c_2$. Both challenges request the same basis. The counterfeiter measures in that basis and sends the same answer to both — success probability 1, since both answers are identical and correct.

Case 2: $c_1 \neq c_2$ (**WLOG** $c_1 = 0, c_2 = 1$). One challenge asks for Z -basis, the other for X -basis. The counterfeiter has a single qubit encoded in one of the two bases and must produce answers for *both* bases. This is where the difficulty lies.

The overall success probability per qubit is:

$$p_{\text{success}} = \frac{1}{2} \underbrace{(1)}_{c_1 = c_2} + \frac{1}{2} \underbrace{(p_{\text{diff}})}_{c_1 \neq c_2},$$

where p_{diff} is the success probability when challenges differ. If we show $p_{\text{diff}} = \frac{1}{2} + \frac{\sqrt{2}}{4}$, then

$$p_{\text{success}} = \frac{1}{2} + \frac{1}{2} \left(\frac{1}{2} + \frac{\sqrt{2}}{4} \right) = \frac{3}{4} + \frac{\sqrt{2}}{8}.$$

9.4 Deriving the bound for differing challenges

When $c_1 = 0$ and $c_2 = 1$, the relevant operator in the SDP (after tracing out the challenge registers) takes the block-diagonal form with blocks indexed by answer pairs (s, t) :

$$Q_{c_1=0, c_2=1} = \frac{1}{2d} \sum_{s,t=0}^{d-1} |s\rangle \langle s|_Y \otimes |t\rangle \langle t|_Z \otimes V_{s,t},$$

where $V_{s,t} = |e_s^0\rangle \langle e_s^0| + |e_t^1\rangle \langle e_t^1|$ acts on the ticket space $\mathcal{X}$.

Computing the eigenvalues of $V_{s,t}$. $V_{s,t}$ is a sum of two rank-1 projectors onto (generally) non-orthogonal states. Let $\gamma = \langle e_s^0 | e_t^1 \rangle$ and work in the 2D subspace spanned by $|e_s^0\rangle$ and $|e_t^1\rangle$.

Write $|e_t^1\rangle = \gamma |e_s^0\rangle + \sqrt{1 - |\gamma|^2} |e_s^\perp\rangle$ for some state $|e_s^\perp\rangle$ orthogonal to $|e_s^0\rangle$. In the basis $\{|e_s^0\rangle, |e_s^\perp\rangle\}$:

$$V_{s,t} = \begin{pmatrix} 1 + |\gamma|^2 & \gamma^* \sqrt{1 - |\gamma|^2} \\ \gamma \sqrt{1 - |\gamma|^2} & 1 - |\gamma|^2 \end{pmatrix}.$$

The trace is 2 and the determinant is $(1 + |\gamma|^2)(1 - |\gamma|^2) - |\gamma|^2(1 - |\gamma|^2) = 1 - |\gamma|^2$. The eigenvalues are:

$$\lambda_{\pm} = \frac{2 \pm \sqrt{4 - 4(1 - |\gamma|^2)}}{2} = 1 \pm |\gamma|.$$

So $V_{s,t}$ has eigenvalues $1 + |\gamma|$ and $1 - |\gamma|$, where $|\gamma| = |\langle e_s^0 | e_t^1 \rangle|$.

Assembling the bound. The dual SDP for the $(c_1 \neq c_2)$ block requires $Y \geq \frac{1}{2d} V_{s,t}$ for all (s, t) . Choosing $Y = \mu \mathbb{1}$ requires $\mu \geq \frac{1}{2d} (1 + |\gamma|)$ for all s, t . The worst case is:

$$\mu = \frac{1 + \sqrt{c}}{2d}, \quad \text{where } c = \max_{s,t} |\langle e_s^0 | e_t^1 \rangle|^2 \text{ (the effective overlap).}$$

The dual objective for this block is $\text{Tr}(Y) = d\mu = \frac{1 + \sqrt{c}}{2}$.

Averaging over all challenge pairs (half have $c_1 = c_2$, half have $c_1 \neq c_2$):

$$p_{\text{success}} \leq \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1 + \sqrt{c}}{2} = \frac{3}{4} + \frac{\sqrt{c}}{4}.$$

For Wiesner's scheme with Z/X bases: $c = |\langle 0 | + \rangle|^2 = 1/2$, giving

$$p_{\text{success}} \leq \frac{3}{4} + \frac{\sqrt{1/2}}{4} = \frac{3}{4} + \frac{\sqrt{2}}{8} \approx 0.927.$$

9.5 Why classical verification is weaker

Compare the two settings:

	Quantum verif.	Classical verif.
Bank receives	physical qubits	classical bits
Bank checks	each qubit state	only matching-basis qubits
Success per qubit	$3/4 = 0.75$	$3/4 + \sqrt{2}/8 \approx 0.927$
Security for $n = 100$	$\sim 10^{-13}$	$\sim 10^{-3}$

The gap arises because in classical verification, half the challenges (those not matching the encoding basis) are “free” for the counterfeiter. In quantum verification, the bank checks *every* qubit directly.

10 Layer 9: Higher-Dimensional Generalizations (Qudits)

10.1 Mutually unbiased bases

Definition 10.1. Two orthonormal bases $\mathcal{B}_0 = \{|e_0^0\rangle, \dots, |e_{d-1}^0\rangle\}$ and $\mathcal{B}_1 = \{|e_0^1\rangle, \dots, |e_{d-1}^1\rangle\}$ of $\mathbb{C}^d$ are *mutually unbiased (MUBs)* if

$$|\langle e_s^0 | e_t^1 \rangle| = \frac{1}{\sqrt{d}} \quad \text{for all } s, t \in \{0, \dots, d-1\}.$$

Interpretation. If a system is prepared in a state from $\mathcal{B}_0$ and measured in $\mathcal{B}_1$, every outcome is equally likely (probability $1/d$). MUBs are the “most incompatible” pair of bases—maximum uncertainty about one given knowledge of the other. For $d = 2$, the Z- and X-bases are MUBs.

10.2 Generalized Pauli operators and the Fourier transform

For a d -dimensional system, define the **shift** and **clock** operators:

$$X_d : |i\rangle \mapsto |i+1 \bmod d\rangle, \quad Z_d : |i\rangle \mapsto \omega^i |i\rangle,$$

where $\omega = e^{2\pi i/d}$ is a primitive d -th root of unity. These satisfy $X_d Z_d = \omega Z_d X_d$ (they q -commute).

The **quantum Fourier transform (QFT)** is:

$$F : |i\rangle \mapsto \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \omega^{ij} |j\rangle.$$

Key property: $F Z_d F^\dagger = X_d$, generalizing $H Z H = X$.

Building MUBs. Set $|e_t^0\rangle = |t\rangle$ (the computational basis) and $|e_t^1\rangle = F |t\rangle$ (the Fourier basis). Then:

$$|\langle e_s^0 | e_t^1 \rangle| = |\langle s | F | t \rangle| = \frac{1}{\sqrt{d}} |\omega^{st}| = \frac{1}{\sqrt{d}} \quad \text{for all } s, t.$$

So the computational and Fourier bases are MUBs, and the effective overlap is $c = 1/d$.

10.3 Results for qudits

Applying the classical-verification framework:

$$p_{\text{success}} = \frac{3}{4} + \frac{\sqrt{c}}{4} = \frac{3}{4} + \frac{1}{4\sqrt{d}}.$$

For $d = 2$: $3/4 + \sqrt{2}/8 \approx 0.927$ (recovers the qubit result). As $d \rightarrow \infty$: approaches $3/4$, showing that higher-dimensional systems improve classical-verification security.

For quantum verification with 3-designs: optimal probability $2/(d+1)$ per qudit. For $d = 2$: $2/3$. For $d = 10$: $2/11 \approx 0.18$.

10.4 The matching lower bound: Schur's lemma and the POVM

Lemma 10.2 (Schur's lemma, representation-theoretic form). *If a set of operators $\{U_g : g \in G\}$ forms an irreducible representation of a group G on $\mathbb{C}^d$, and M commutes with all U_g , then $M = \lambda \mathbb{1}$ for some scalar λ .*

The paper constructs the counterfeiting POVM as follows. Define:

$$|\psi\rangle = (2 + 2/\sqrt{d})^{-1/2}(|0\rangle + F|0\rangle),$$

and for each $(s, t) \in \{0, \dots, d-1\}^2$:

$$P_{s,t} = (X_d^s Z_d^t) |\psi\rangle \langle \psi| (X_d^s Z_d^t)^\dagger.$$

The operators $\{X_d^s Z_d^t : s, t \in \mathbb{Z}_d\}$ form the **Heisenberg–Weyl group** (up to phases), which acts irreducibly on $\mathbb{C}^d$. The operator $M = \frac{1}{d} \sum_{s,t} P_{s,t}$ commutes with all elements of this group, so by Schur's lemma, $M = \lambda \mathbb{1}$. Taking the trace: $\text{Tr}(M) = d^2 \cdot \frac{1}{d} \cdot 1 = d = \lambda \cdot d$, so $\lambda = 1$ and $\{P_{s,t}/d\}$ is a valid POVM.

The attack: on challenge pair (c_1, c_2) with $c_1 \neq c_2$, measure with POVM $\{P_{s,t}/d\}$ and return s for basis-0, t for basis-1. The success probability computation (in the paper) shows this achieves $\frac{1}{2}(1 + 1/\sqrt{d})$ on the differing-challenge case, yielding overall $3/4 + 1/(4\sqrt{d})$, matching the dual upper bound.

11 Layer 10: Connections and Broader Context

11.1 Connection to quantum cloning

The relationship between quantum money and quantum cloning is:

Cloning concept		Money concept
Universal cloner (all states)	$\leftrightarrow$	Scheme with 3-design ensemble
State-specific cloner	$\leftrightarrow$	Scheme with specific ensemble
Joint fidelity $\langle \psi\psi \rho_{\text{out}} \psi\psi \rangle$	$\leftrightarrow$	Counterfeiting probability
Bužek–Hillery cloner	$=$	Optimal attack on 6-state scheme
Phase-covariant cloner	$\sim$	Optimal attack on 4-state scheme
Werner's cloner (d -dim)	$=$	Optimal attack on 3-design scheme

11.2 Connection to BB84

Wiesner’s conjugate coding is the direct ancestor of BB84 quantum key distribution. Both use the same $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$ states and the same information-disturbance principle. The connection:

	Quantum money	BB84 QKD
Secret	bank’s key k	shared key
Adversary	counterfeiter	eavesdropper
Adversary’s task	clone the state	intercept & resend
Detection	verification fails	error rate rises
Security from	no-cloning + SDP	no-cloning + privacy amplification

11.3 Connection to uncertainty relations

The effective overlap $c = \max_{s,t} |\langle e_s^0 | e_t^1 \rangle|^2$ is directly connected to **entropic uncertainty relations**. For two measurements with bases $\mathcal{B}_0, \mathcal{B}_1$:

$$H(\mathcal{B}_0) + H(\mathcal{B}_1) \geq -2 \log \sqrt{c},$$

where $H(\mathcal{B})$ is the Shannon entropy of the measurement outcomes. This is the Maassen–Uffink uncertainty relation. Smaller c (more unbiased bases) means more uncertainty, which means harder counterfeiting. MUBs ($c = 1/d$) maximize this lower bound.

11.4 Connection to state discrimination

The classical-verification counterfeiting problem is a special case of **quantum state discrimination**. For each answer pair (a_1, a_2) , define the mixed state $\rho_{a_1 a_2} = \sum_k w_k |\psi_k\rangle \langle \psi_k|$ (weighted over keys k for which (a_1, a_2) is a valid answer to some challenge pair). The counterfeiter must determine which “answer hypothesis” is correct by measuring the ticket state. The SDP for optimal state discrimination (Eldar–Megretski–Verghese 2003) is the same as the counterfeiting SDP.

11.5 Open directions

The paper analyzes “simple counterfeiting attacks” ($1 \text{ note} \rightarrow 2 \text{ copies}$). Natural extensions include:

1. **Adaptive attacks:** The counterfeiter interacts with the bank’s verification oracle multiple times. Pastawski et al. showed that if the oracle only returns accept/reject, adaptive access does not help much.
2. $m \rightarrow m + 1$ **counterfeiting:** Given m notes, produce $m + 1$. Aaronson–Christiano showed this reduces to the $1 \rightarrow 2$ case.
3. **Public-key quantum money:** Verification without any bank communication, relying on computational hardness (e.g., Aaronson’s scheme based on hidden subspaces). This requires complexity-theoretic assumptions rather than information-theoretic security.
4. **Quantum lightning:** A strengthening where even the bank cannot produce two copies of the same note (useful for preventing double-spending).

12 Summary of Main Results

Scheme	Verification	Optimal attack per qubit/qudit
Wiesner 4-state	Quantum	$3/4 = 0.750$
6-state / SIC-POVM	Quantum	$2/3 \approx 0.667$
Optimal d -dim (3-design)	Quantum	$2/(d+1)$
Wiesner classical	Classical	$3/4 + \sqrt{2}/8 \approx 0.927$
Qudit classical (d -dim, MUB)	Classical	$3/4 + 1/(4\sqrt{d})$

For n qubits/qudits, each probability is raised to the n -th power.

Proof technique. Every bound is established by:

1. Formulating the optimal counterfeiting attack as an SDP.
2. Exhibiting a **dual feasible solution** (typically $Y = \|Q\| \mathbb{1}$) to obtain an upper bound.
3. Exhibiting a **primal feasible solution** (an explicit channel or POVM) achieving the same value.
4. Concluding by strong duality that both are optimal.